

Assignment: K-Nearest Neighbors (KNN) Implementation

1. Build a Full KNN-Based Classification Workflow

You must construct a comprehensive machine learning pipeline that includes:

- Project hosting on GitHub.
- Automatic data retrieval via code.
- Data preprocessing.
- Model training using the K-Nearest Neighbors (KNN) algorithm.
- Evaluation and visualization of the model's performance.

2. Perform Essential Data Preprocessing

Since KNN is a distance-based algorithm, preprocessing is strictly required. Your pipeline must:

- Handle missing values (Imputation or removal).
- Encode categorical features (One-Hot Encoding or Label Encoding).
- Split data: Create Train/Test/Validation sets (e.g., 80/20 or 70/15/15).

3. GitHub Repository Setup (Crucial Step)

Instead of using Google Drive, you must host your project on GitHub:

- Create a new *public* GitHub repository.
- Upload your selected dataset (in CSV) to this repository.
- Upload your Google Colab Notebook (.ipynb) to this repository.
- Requirement: The repository must contain all necessary components to run the project.

4. Model Implementation in Google Colab

Your notebook must demonstrate the following steps:

- Data Loading (Automation): You must not require the user to manually upload a file.
- Preprocessing: Apply the steps outlined in Section 2.
- KNN Model Training: Initialize and fit the classifier.
- Hyperparameter Tuning: Use techniques like Grid Search or Cross-Validation to find the optimal:
 - k (Number of neighbors)
 - Weights (uniform vs distance)
 - Metric (e.g., minkowski, euclidean, manhattan)
- Evaluation: Generate metrics on the test set.

5. Include the Following Screenshots

Capture and include the following visuals in your submission:

- Confusion Matrix: A heatmap visualization of True Positives, False Positives, etc.
- Elbow Method Plot (or Error Rate vs. K-Value): A line graph showing how accuracy/error changes as the value of k increases, highlighting the optimal k.
- Evaluation Metrics: Accuracy, Precision, Recall, F1-score, and AUC.
- ROC Curve: Plot of the True Positive Rate vs. False Positive Rate.
- 2D Decision Boundary Plot: A visualization of the classification regions

6. Submission

- GitHub Link: The URL to your public repository containing both the dataset and the .ipynb file.
- Colab Link: The direct link to your notebook
- **I should be able to click the Colab link, run all cells, and have the dataset download and the model train automatically without manual intervention, with plots generated.**